

Homework 9

Problem 1. Use the Lovasz local lemma to show that if

$$4 \binom{k}{2} \binom{n}{k-2} 2^{1-\binom{k}{2}} \leq 1$$

then it is possible to color the edges of K_n with two colors so that it has no monochromatic (i.e. single color) K_k subgraph.

Solution. E_i : the i -th K_k is monochromatic. $Pr(E_i) = 2^{1-\binom{k}{2}}$. Consider the dependency graph, for any different E_i and E_j , they are adjacent if the corresponding K_k share at least one edge. Thus the degree of the dependency graph is bounded by $\binom{k}{2} \binom{n}{k-2}$.

According to the Lovasz local lemma, it is possible that none of the E_i happens under the given inequality. $\square$

Problem 2. What is the expected number of trees with k vertices in $G \in \mathcal{G}(n, p)$?

Solution. By Cayley's formula and the linearity of expectation, it is $\binom{n}{k} k^{k-2} p^{k-1}$. $\square$

Problem 3. suppose that $p = \frac{d}{n}$ where $d = o(n^{1/3})$. Show that $G \in \mathcal{G}(n, p)$ almost surely has no copies of K_4 .

Solution. The probability that there exists some 4-vertices sub-graph which induces a K_4 is upper bounded by $p = \binom{n}{4} p^6$. When $d = o(n^{1/3})$, $\lim_{n \rightarrow \infty} p = 0$. $\square$

Problem 4. Show that if almost all $G \in \mathcal{G}(n, p)$ have a graph property $\mathcal{P}_1$ and almost all $G \in \mathcal{G}(n, p)$ have a graph property $\mathcal{P}_2$, then almost all $G \in \mathcal{G}(n, p)$ have both properties.

Solution. The portion of the graphs have both properties equals 1 minus the portion of the graphs which does not have property $\mathcal{P}_1$ or $\mathcal{P}_2$. However the portion of the graph does not have property $\mathcal{P}_1$ or $\mathcal{P}_2$ is bounded by the sum of the portion of the graphs does not have property $\mathcal{P}_1$ and the the portion of the graphs does not have property $\mathcal{P}_2$, which both tend to 0 as n approaches ∞ . The claim in the question then follows. $\square$